

Inconsistencies of interannual variability and trends in long-term satellite leaf area index products

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Abstract

Understanding the long-term performance of global satellite leaf area index (LAI) products is important for global change research. However, few effort has been devoted to evaluating the long-term time-series consistencies of LAI products. This study compared four long-term LAI products (GLASS, GLOBMAP, LAI3g, and TCDR) in terms of trends, interannual variabilities, and uncertainty variations from 1982 through 2011. This study also used four ancillary LAI products (GEOV1, MERIS, MODIS C5, and MODIS C6) from 2003 through 2011 to help clarify the performances of the four long-term LAI products. In general, there were marked discrepancies between the four long-term LAI products. During the pre-MODIS period (1982–1999), both linear trends and interannual variabilities of global mean LAI followed the order GLASS>LAI3g>TCDR>GLOBMAP. The GLASS linear trend and interannual variability were almost 4.5 times those of GLOBMAP. During the overlap period (2003–2011), GLASS and GLOBMAP exhibited a decreasing trend, TCDR no trend, and LAI3g an increasing trend. GEOV1, MERIS, and MODIS C6 also exhibited an increasing trend, but to a much smaller extent than that from LAI3g. During both periods, the R^2 of detrended anomalies between the four long-term LAI products was smaller than 0.4 for most regions. Interannual variabilities of the four long-term LAI products were considerably different over the two periods, and the differences followed the order GLASS>LAI3g>TCDR>GLOBMAP. Uncertainty variations quantified by a collocation error model followed the same order. Our results indicate that the four long-term LAI products were neither intraconsistent over time nor interconsistent with each other. These inconsistencies may be due to NOAA satellite orbit changes and MODIS sensor degradation. Caution should be used in the interpretation of global changes derived from the four long-term LAI products.

KEYWORDS

GLASS, GLOBMAP, intercomparison, LAI3g, leaf area index (LAI), TCDR, time series

1 | INTRODUCTION

Leaf area index (LAI), indicating the area of green leaves per unit of ground surface area, is identified as an essential climate variable

(ECV) by the global climate observing system (GCOS) (Gobron & Verstraete, 2009). The acquisition of reliable global LAI data is a prerequisite for modeling, mapping, and monitoring land surface ecosystems (Asner, Scurlock, & Hicke, 2003; Ryu et al., 2011; Sellers et al.,

1995). Recently, several long-term (≥ 30 years) global LAI products have been produced (Claverie, Matthews, Vermote, & Justice, 2016; Liu, Liu, & Chen, 2012; Xiao et al., 2016; Zhu et al., 2013). These long-term products are of great importance in understanding the responses and feedback of land surface ecosystems to climate changes, elevated CO₂ concentrations, and anthropogenic activities (Mao et al., 2016; Piao et al., 2015; Zhu et al., 2016).

Appropriate use of global satellite LAI products requires comprehensive evaluation studies. To address this issue, the Committee on Earth Observation Satellites (CEOS) Land Product Validation (LPV) subgroup proposed a four-stage validation hierarchy (Morissette et al., 2006). Under this framework, both theoretical and physical uncertainties of global satellite LAI products have been investigated (Fang, Wei, Jiang, Scipal, 2012; Fang, Wei, Liang, 2012) against a target accuracy requirements of $\pm 20\%$ proposed by GCOS (GCOS, 2011). However, most existing evaluation studies focused on fine-resolution (≤ 1 km) and short-term (e.g., since the 2000s) LAI products, in terms of absolute magnitude (Garrigues et al., 2008), spatial pattern (Fang et al., 2013), and seasonal trajectory (Verger, Filella, Baret, & Peñuelas, 2016). In comparison, less effort has been devoted from a long-term perspective (i.e., since the 1980s).

Long-term global satellite LAI products are derived from advanced very high-resolution radiometer (AVHRR) datasets on board of the national oceanic and atmospheric administration (NOAA) satellite series, which are prone to larger uncertainties than short-term datasets (Guay et al., 2014; Tian et al., 2015). Sources of uncertainty in AVHRR time-series data mainly involve sensor degradation, lack of onboard calibration, drift of satellite orbits, and changes in platforms and sensors (Pinzon & Tucker, 2014; Tucker et al., 2005). These data source uncertainties are likely to propagate to LAI products, along with additional uncertainties caused by algorithms. Platform-related uncertainties may vary gradually or abruptly and, therefore, the performances of LAI products and other downstream products, such as gross primary productivity (GPP), may also vary over time. To meet the needs of global climate studies, GCOS has proposed guidelines requiring a stability of $\pm 10\%$ for LAI products (GCOS, 2011). Therefore, evaluating global satellite LAI products from a long-term perspective is of great importance to ensure their correct application in global change research.

Trends and interannual variabilities in terrestrial ecosystems are two overarching issues that long-term LAI products should address. Many studies have reported a greening earth related to global warming and elevated CO₂ concentration (Fensholt & Proud, 2012; Myneni, Keeling, Tucker, Asrar, Nemani, 1997; Myneni, Nemani, Running, 1997) and oscillations caused by El Niño and other extreme events (de Jong, Verbesselt, Schaepman, & de Bruin, 2012; Myneni, Tucker, Asrar, & Keeling, 1998), based on analyses of long-term normalized difference vegetation index (NDVI) products. LAI has an advantage over NDVI due to its explicit ecological meaning, so that global long-term satellite LAI products have been considered as references to interpret LAI time series simulated by land surface models (Lucht et al., 2002; Mao et al., 2013; Piao, Friedlingstein, Ciais, Zhou, & Chen, 2006). However, the degrees

of trends and interannual variabilities are uncertain because the agreements and discrepancies between different long-term LAI products remain unclear.

This study was performed to undertake a detailed evaluation on the consistencies of four long-term global satellite LAI products (Table 1): Global Land Surface Satellite (GLASS) (Xiao et al., 2016); GLOBMAP (Liu et al., 2012); LAI3g (Zhu et al., 2013); and terrestrial climate data record (TCDR) (Claverie et al., 2016). Aiming at this goal, we examined their trends, interannual variabilities, and uncertainty variations. We also compared the four long-term LAI products with other widely used but short-term products: GEOV1 (Baret et al., 2013); MERIS (Tum et al., 2016); MODIS Collection 5 (C5) (Shabanov et al., 2005); and MODIS Collection 6 (C6) (Yan et al., 2016). Consequently, three time periods were investigated separately: 30-year long-term period (1982–2011), pre-MODIS period (1982–1999), and overlap period (2003–2011). We address a single question: are current long-term LAI products intraconsistent (having comparable spatiotemporal patterns itself over different time periods) and interconsistent (having comparable spatiotemporal patterns with each other)?

2 | MATERIALS AND METHODS

2.1 | Long-term LAI products from 1982–2015

2.1.1 | GLASS LAI product

GLASS product (version 03), acquired from <ftp://ftp.glc.fumd.edu/>, is generated every 8 days, using 0.05° resolution NOAA/AVHRR surface reflectance datasets provided by NASA's Long Term Data Record (LTDR) project before 2001 (Pedelty et al., 2007), and 1 km resolution Terra/MODIS surface reflectance datasets (MOD09) since 2001 (Xiao et al., 2016). The algorithm fuses Terra/MODIS LAI (MOD15) with clump-corrected CYCLOPES LAI, and establishes relationships between the fused LAI dataset and NOAA/AVHRR or Terra/MODIS surface reflectance over Benchmark Land Multisite Analysis and Intercomparison of Products (BELMANIP) sites using biome-specific general regression neural networks (Xiao et al., 2016). Yearly reflectance–LAI pairs are used to train the neural networks and estimate LAI time series (Xiao et al., 2014).

2.1.2 | GLOBMAP LAI product

GLOBMAP product (version 01), acquired from <http://www.globalmapping.org/>, is generated in 1/13.75° spatial resolution, using biweekly NDVI provided by the global inventory modeling and mapping studies (GIMMS) before March 2000 (Tucker et al., 2005), and 8-day Terra/MODIS surface reflectance (MOD09) since March 2000. The algorithm inverses a geometrical optical model to generate MODIS LAI from MOD09 data since March 2000, and then establishes pixel-based relationships between MODIS LAI and simple ratio (SR) vegetation index derived from GIMMS NDVI during the overlapping period from 2000 to 2006. Such relationships are finally used to derive LAI from AVHRR SR before March 2000 (Liu et al., 2012).

TABLE 1 Summary of global satellite LAI products

Product	Time span ^a	Data source	Spatial resolution	Temporal resolution	References
GLASS	1982–2014	NOAA/AVHRR LTDR reflectance & Terra/MODIS C5 reflectance	0.05°, 1982–1999 1 km, 2001–2011	8-day	(Xiao et al., 2016)
GLOBMAP	1982–2011	GIMMS NDVI & Terra/MODIS C5 reflectance	1/13.75°	Biweekly, 1982–1999/2 8-day, 2000/3–2011	(Liu et al., 2012)
LAI3g	1982–2011	GIMMS NDVI3g	1/12°	Biweekly	(Zhu et al., 2013)
TCDR	1982–2015	NOAA/AVHRR TCDR reflectance	0.05°	Daily	(Claverie et al., 2016)
GEOV1	1999–2013	SPOT/VGT reflectance	1/112°	10-day	(Baret et al., 2013)
MERIS	2003–2011	ENVISAT/MERIS reflectance	1/360°	10-day	(Tum et al., 2016)
MODIS C5	2001–2015	Terra/MODIS C5 reflectance	1 km	8-day	(Shabanov et al., 2005)
MODIS C6	2001–2015	Terra/MODIS C6 reflectance	500 m	8-day	(Yan et al., 2016)

^aYears with year-round data.

2.1.3 | LAI3g LAI product

LAI3g product (version 01), acquired from <http://sites.bu.edu/clive/g/datacodes/>, is generated biweekly in 1/12° spatial resolution (Zhu et al., 2013). The input data are the third-generation NDVI (NDVI3g) provided by GIMMS. The algorithm is based on feed-forward neural networks connecting the MODIS LAI Beijing Normal University (BNU) version (Yuan, Dai, Xiao, Ji, & Shangguan, 2011) to NDVI3g. This calibration procedure is conducted on multiyear average monthly LAI and NDVI data, so that 12 neural networks are involved, one for each month. The calibrated neural networks are further used to retrieve AVHRR LAI from NDVI3g.

2.1.4 | TCDR product

AVHRR TCDR product (version 04), acquired from <ftp://eclipse.ncdc.noaa.gov/>, is provided every day in 0.05° spatial resolution (Claverie et al., 2016). The product is mainly derived from a surface reflectance dataset obtained by NOAA/AVHRR sensors, which has been geolocated, calibrated, cloud and snow removed, and corrected for atmospheric and bidirectional reflectance distribution function (BRDF) effects (Vermote, Justice, & Bréon, 2009). The algorithm is based on an artificial neural network connecting Terra + Aqua combined MODIS LAI (MCD15) to AVHRR TCDR reflectance separately for five biomes (Claverie et al., 2016). The calibration procedure is conducted using data over BELMANIP2 sites, and the calibrated neural network is further used to retrieve AVHRR LAI from TCDR reflectance.

2.2 | Ancillary LAI products from 1999–2015

2.2.1 | GEOV1 LAI product

GEOV1 LAI product (version 1.3), acquired from <http://land.copernicus.eu/>, is generated every 10 days in 1/112° spatial resolution. MODIS and CYCLOPES LAI products are fused to generate the “best estimates” of LAI using a linear formula (Baret et al., 2013). A neural network is trained between the fused LAI and the SPOT/VEGETATION top of canopy directionally normalized reflectance values over the global BELMANIP sites (Baret et al., 2006). The trained neural

network provides LAI estimates from the SPOT/VEGETATION reflectance.

2.2.2 | MERIS LAI product

MERIS LAI product, acquired from <https://centaurus.caf.dlr.de:8443/>, is generated every 10 days in 1/360° spatial resolution (Tum et al., 2016). The algorithm is based on the training of a neural network over a database of simulated top-of-atmosphere radiances, using a coupled leaf-canopy-atmosphere radiative transfer model (Bacour, Baret, Béal, Weiss, & Pavageau, 2006). The trained neural network is used to estimate LAI from ENVISAT/MERIS L1B observations at 15 spectral bands. A harmonic analysis is further incorporated into the algorithm to fill data gaps caused by irregular sampling strategy of MERIS full-resolution (FR) data.

2.2.3 | MODIS C5 and C6 LAI products

MODIS LAI products, acquired from <ftp://ladsweb.nascom.nasa.gov/>, are generated every 8 days in 1 km and 500 m spatial resolution, for MOD15A2 C5 and MOD15A2H C6, respectively. The main algorithm employs biome-specific look-up tables (LUTs) simulated from a 3D radiative transfer model to search for LAI values for a specific set of sun-object-sensor geometries, and observed red and near-infrared reflectances from Terra/MODIS C5 and C6 reflectance, respectively (Huang et al., 2008; Knyazikhin, Martonchik, Myneni, Diner, & Running, 1998; Yan et al., 2016). When the main algorithm fails, a backup algorithm based on the LAI–NDVI relationships derived from model simulations is used for LAI estimation (Myneni, Keeling, et al., 1997; Myneni, Nemani, et al., 1997). A major improvement in MODIS C6 LAI product is the usage of MOD09GA C6 reflectance product, which applies a new sensor calibration.

2.3 | Data processing and analysis

2.3.1 | Calculation of global mean LAI

We obtained annual LAI maps and global mean LAI time series from each LAI product. Monthly mean 0.5° LAI was calculated by

averaging LAI values from original spatiotemporal resolution. The residual data gaps were filled by a widely used Harmonic Analysis of Time Series (HANTS) method, which applies a least squares curve fitting procedure for time-series signals based on harmonic components of periodic functions (Roerink, Menenti, & Verhoef, 2000; Zhou, Jia, & Menenti, 2015). We calculated annual growing season mean LAI maps by averaging monthly LAI values with monthly mean temperature higher than 0°C (Fig. S1) (Prentice et al., 2011). Monthly averaged daily mean temperatures (1982–2015) were obtained from the climatic research unit (CRU) high-resolution gridded datasets (version 3.24) (Harris, Jones, Osborn, & Lister, 2014), and we calculated multiyear averaged monthly mean temperatures to ensure that the same growing season land mask was used over the entire period (1982–2015) across different products (Fig. S2). The global mean LAI values were calculated by area-weighted average over annual growing season mean LAI maps excluding unvegetated regions (Fig. S3).

2.3.2 | Time-series analysis

We used an ensemble empirical mode decomposition (EEMD) method along with the classical linear model method to detect LAI trend. The EEMD method decomposes time series into a group of oscillatory components at different frequency levels without any predetermined basis functions (Huang et al., 1998), and EEMD is its enhanced version overcoming the scale mixing problem (Wang, 2009). The number of components usually varies from three to eight, depending on the length of the time series (Huang et al., 2003). In this study, we decomposed the long-term and ancillary LAI products into four and three components, respectively. Summing different components generates adaptive trends at different temporal scales (Ji, Wu, Huang, & Chassignet, 2014). We summed the last two components to account for decadal-scale trends according to Wu, Huang, Long, and Peng (2007). Annual anomalies were obtained by subtracting adaptive trends from the original time series, and the standard deviation of each detrended anomaly was calculated as a quantitative metric of the interannual variability. Linear trends and interannual variabilities were also calculated pixel by pixel to generate spatial patterns.

We used a collocation error model to quantify uncertainty time series of LAI products. The triple collocation error model (TCEM) is the primary implementation of the collocation model, which obtains analytical solutions of random errors of three spatially and temporally collocated remote sensing products with no need for a “true” observation (Gruber et al., 2016; Stoffelen, 1998). We extended the original TCEM to a general collocation error model (GCEM) so that more than three collocated datasets can be involved. For m collocated datasets, $\mathbf{x}_i$ ($i = 1, 2, \dots, m$), the covariance between each pair of datasets, including intrapair cases, is given by (McColl et al., 2014):

$$\text{Cov}(\mathbf{x}_i, \mathbf{x}_j) = \theta_i \theta_j + \text{Cov}(\varepsilon_i, \varepsilon_j) \quad (1)$$

where ε_i is the random error of $\mathbf{x}_i$, θ_i is a parameter related to system error of $\mathbf{x}_i$ and the variation of hypothetical truth, and $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$

when $i \neq j$ according to the independent error assumption (McColl et al., 2014). A total of $m(m+1)/2$ equations can be obtained, along with $2m$ unknown parameters, i.e., θ_i and σ_i ($i = 1, 2, \dots, m$), where $\sigma_i^2 = \text{Cov}(\varepsilon_i, \varepsilon_i)$ and σ_i is taken as uncertainty. When $m \geq 3$ GCEM is a well-constrained and solvable system, and a numerical solution can be achieved using the least squares technique (Pan et al., 2015). In this study, pixels over one land cover type in a continent constituted sampling of $\mathbf{x}_i$ and yielded one regional uncertainty estimate. The land cover type was derived from the global land cover product for the year 2000 (GLC2000) (Bartholomé & Belward, 2005; Giri, Zhu, & Reed, 2005) (Fig. S3). Regions with less than 60 pixels were masked to avoid numerical problems caused by inadequate sampling of $\mathbf{x}_i$ (Fang, Wei, Jiang, et al. 2012; Fang, Wei, Liang, 2012). Global LAI uncertainty was calculated yearly as the sum of regional uncertainties, weighted by the area of each region. We used a 2×standard deviation of each uncertainty time series, corresponding to 95% range of variations assuming an approximately normal distribution, as the quantitative metric of the temporal uncertainty variation. Relative uncertainty and relative uncertainty variation were also calculated, using LAI value as a normalization factor according to GCOS (2011).

3 | RESULTS

3.1 | Linear trends

Global mean LAI values generally varied within the range of 1.55 ± 0.3 according to the four long-term and four ancillary LAI products (Figure 1). The long-term products showed higher global mean LAI values than ancillary products after 2001, with GLOBMAP as an exception that showed the lowest values. The largest differences appeared in 2011, when LAI3g was 45% higher than GLOBMAP, 30% higher than MODIS C5, and 25% higher than MODIS C6. The global mean values for TCDR, GLASS, and LAI3g differed from each other before 1995, but became similar after 2001.

The linear trends of the four long-term LAI products varied substantially (Figure 1 and Table 2). Over the 30 years (1982–2011), GLASS, LAI3g, and TCDR showed significant increasing trends, while GLOBMAP showed a significant decreasing trend. The mean trend of these four products was about $0.053 \text{ m}^2 \text{ m}^{-2} \text{ decade}^{-1}$, with a coefficient of variation of 119%. Particularly, the difference between GLASS and GLOBMAP reached $0.154 \text{ m}^2 \text{ m}^{-2} \text{ decade}^{-1}$. There were notable differences between 1982–1999 and 2003–2007. GLASS and GLOBMAP changed from significant increasing trends (0.115 and $0.026 \text{ m}^2 \text{ m}^{-2} \text{ decade}^{-1}$, respectively) to significant decreasing trends (-0.044 and $-0.098 \text{ m}^2 \text{ m}^{-2} \text{ decade}^{-1}$, respectively). LAI3g maintained a significant increasing trend, but its slope value almost doubled ($0.082 \text{ m}^2 \text{ m}^{-2} \text{ decade}^{-1}$ for 1982–1999 and 0.151 for 2003–2007). TCDR changed from a significant increasing trend ($0.057 \text{ m}^2 \text{ m}^{-2} \text{ decade}^{-1}$) during 1982–1999 to an insignificant trend ($0.004 \text{ m}^2 \text{ m}^{-2} \text{ decade}^{-1}$) during 2003–2007. While the four long-term LAI products disagreed in trend signs and values during the overlap period (2003–2011), the three ancillary LAI products, GEOV1, MERIS, and MODIS C6, agreed very well, with a

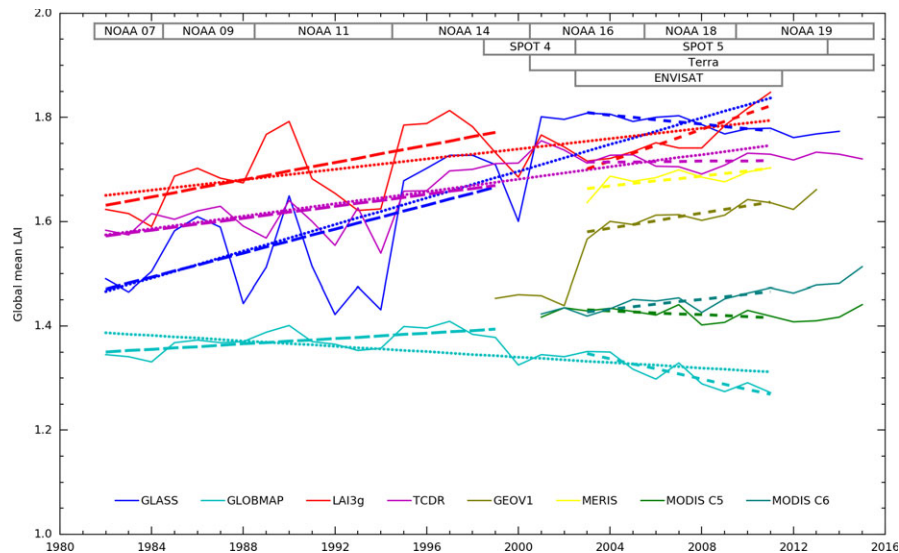


FIGURE 1 Global mean LAI (solid curves) and linear trends during 1982–2011 (dotted lines), 1982–1999 (long dashed lines), and 2003–2011 (short dashed lines) for different LAI products. Trend values are listed in Table 2. Lifespans of different satellite platforms are also illustrated [Colour figure can be viewed at wileyonlinelibrary.com]

mean trend ($0.056 \text{ m}^2 \text{ m}^{-2} \text{ decade}^{-1} \pm 2 \text{ SD}$) ranged about $0.028 \sim 0.084 \text{ m}^2 \text{ m}^{-2} \text{ decade}^{-1}$ (Figure 1 and Table 2). However, the trend values from the four long-term LAI products all fell far out of this range. It is also noted that MODIS C5 had a small decreasing trend ($p > .05$), whereas MODIS C6 had an increasing trend ($p < .05$).

Spatial patterns of linear trends of the four long-term LAI product showed some similarities during 1982–1999 (Figure 2a–d). GLASS, LAI3g, and TCDR revealed a greening earth as around 45% of vegetated land surface area showed an increasing trend, especially in the northern high latitudes. GLOBMAP also showed an increasing trend over those areas. However, it is notable that agreements among these products were limited in trend signs. GLASS exhibited the largest trend values, followed by LAI3g and TCDR, while GLOBMAP exhibited much smaller trends.

Spatial patterns of linear trends of the four long-term LAI product showed large discrepancies during 2003–2011 (Figure 2e–h). More than 20% GLASS and GLOBMAP but only <10% LAI3g and TCDR showed browning of the vegetated land surface in tropical forests, North America, and Central Russia. Only about 10% GLASS, GLOBMAP, and TCDR, but 33% LAI3g showed a greening tendency in the Northern Hemisphere. On contrast, the ancillary GEOV1, MERIS, and MODIS C6 showed quite similar values, with 15%–20% positive trend and about 5% negative. Regions with growing LAI included Eastern Siberia, Europe, Eastern China, India, Eastern Australia, and Southern Africa.

3.2 | Adaptive trends, detrended anomalies, and interannual variabilities

The EEMD method detected decadal-scale adaptive trends and various temporal patterns were demonstrated for the four long-term LAI

products over the three decades examined (Figure 3a). In the first decade (1982–1991), both GLOBMAP and LAI3g showed slight increasing trends, GLASS showed a slight decrease trend, and TCDR showed a flat tendency. During the second decade (1992–2001), GLASS, TCDR, and LAI3g showed overall increasing trend to different degrees, whereas GLOBMAP shows a flat tendency. During the third decade (2002–2011), both GLOBMAP and GLASS showed decreasing trends, LAI3g showed an increasing trend, and TCDR showed a flat tendency. The three ancillary LAI products, GEOV1, MERIS, and MODIS C6, all showed increasing trends, but a clear turning point was observed for GEOV1 around 2003 when its platform changed from SPOT4 to SPOT5. In contrast to MODIS C6, MODIS C5 showed an overall flat tendency.

Diverse patterns of global mean LAI anomalies were observed after removing the adaptive trends (Figure 3b and Table 2). The most distinct feature was the presence of a dividing point around 2001. Before 2001, there were considerable oscillations in anomalies of global mean LAI curves, and the amplitudes differed from each other. During the pre-MODIS period (1982–1999), the mean interannual variability in the four long-term LAI products was $0.044 \text{ m}^2 \text{ m}^{-2} \text{ year}^{-1}$, with a coefficient of variation of 61%. Particularly, the interannual variability of GLASS was 4.4 times that of GLOBMAP. During 2003–2011, the four long-term LAI products showed much smaller variations than those before 2001, with a mean $\pm \text{SD}$ about $0.013 \pm 0.001 \text{ m}^2 \text{ m}^{-2} \text{ year}^{-1}$. The four ancillary LAI products also showed a small mean interannual variability of $0.010 \text{ m}^2 \text{ m}^{-2} \text{ year}^{-1}$ with a standard deviation of $0.003 \text{ m}^2 \text{ m}^{-2} \text{ year}^{-1}$, both of which were comparable with those of the four long-term LAI products.

Different LAI products showed different spatial patterns of interannual variability during 1982–1999 (Figure 4a–d). The tropical forests showed markedly large discrepancies. GLASS, LAI3g, TCDR, and GLOBMAP exhibited >0.20 , 0.15 – 0.20 , 0.10 – 0.15 , and

TABLE 2 Global linear trends, interannual variabilities, uncertainties, and relative uncertainties for different LAI products during different time periods. Bold type indicates a significance level of $p < .05$ for linear trends. SD denotes standard deviation

Product	Linear trend ($\text{m}^2 \text{m}^{-2} \text{decade}^{-1}$)			Interannual variability ($\text{m}^2 \text{m}^{-2} \text{year}^{-1}$)			Relative uncertainty (%)			Relative uncertainty variation (%) 1982–2011
	1982–2011	1982–1999	2003–2011	1982–2011	1982–1999	2003–2011	1982–2011	1982–1999	2003–2011	
GLASS	0.128	0.115	−0.044	0.064	0.074	0.011	11.7	13.3	8.8	3.5
GLOBMAP	−0.026	0.026	−0.098	0.017	0.017	0.014	19.6	19.6	19.5	1.1
LAI3g	0.049	0.082	0.151	0.047	0.057	0.013	8.5	8.1	9.3	2.0
TCDR	0.059	0.057	0.004	0.023	0.027	0.013	13.0	13.6	12.1	1.6
Average	0.053	0.070	0.003	0.038	0.044	0.013	13.2	13.6	12.4	2.1
SD	0.063	0.038	0.107	0.022	0.027	0.001	4.7	4.7	4.9	0.9
GEOV1			0.072			0.013			6.4	
MERIS			0.049			0.010			14.0	
MODIS C6			0.048			0.008			13.6	
Average			0.056			0.010			11.4	
SD			0.014			0.003			4.3	
MODIS C5			−0.019			0.011				

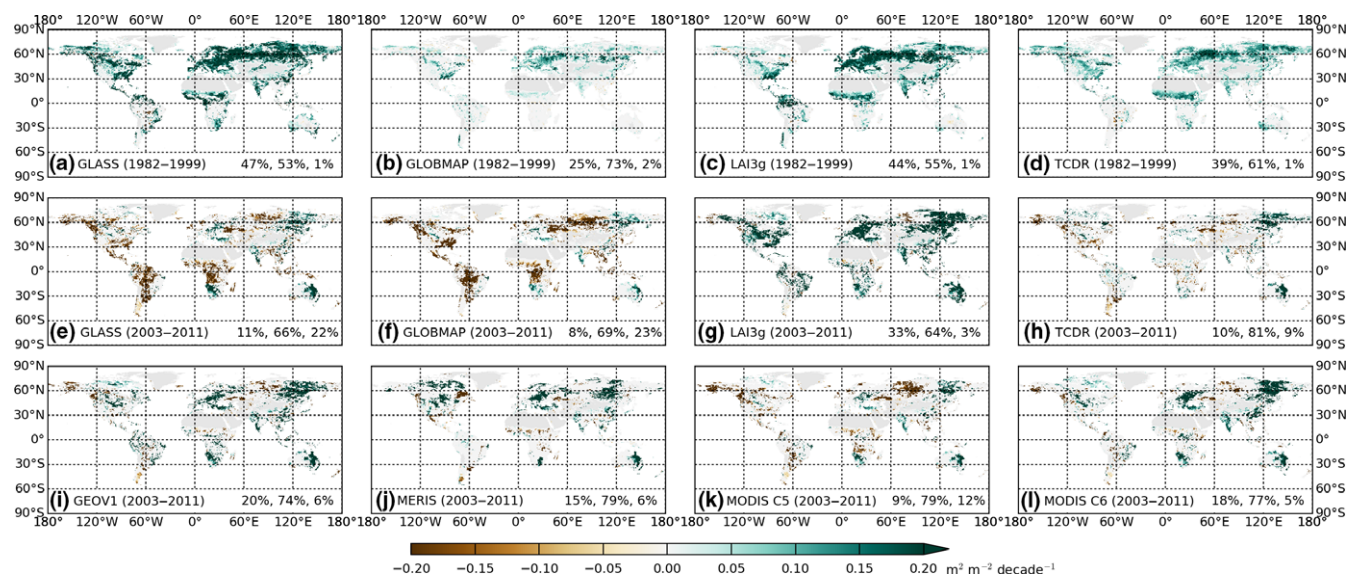


FIGURE 2 Linear trend maps of the growing season mean LAI for different LAI products during different time periods. Trends with $p \geq .1$ were assigned a value of 0 for illustrative purposes. The three numbers in each subplot indicate the area percentages of positive, zero, and negative trends on vegetated land surface [Colour figure can be viewed at wileyonlinelibrary.com]

$<0.15 \text{ m}^2 \text{m}^{-2} \text{year}^{-1}$ interannual variability values in most tropical forest areas, respectively. In temperate forests similar discrepancies were observed, that GLASS exhibited the largest interannual variabilities, followed by LAI3g and TCDR, while GLOBMAP exhibited much smaller interannual variabilities.

Spatial patterns of interannual variabilities of the four long-term LAI product showed some similarities during 2003–2011 (Figure 4e–h). Hotspots were usually observed in tropical forests, Eastern South America, Central Eurasia, Eastern Siberia, and Eastern Australia. These regions were also highlighted in three ancillary LAI products, GEOV1, MODIS C5, and MODIS C6.

3.3 | Uncertainties and changes in uncertainty

Figure 5 showed the global uncertainties and relative uncertainties for the four long-term LAI products and three ancillary LAI products. MODIS C5 was excluded because its error was likely to be highly correlated with MODIS C6, violating the assumption of the collocation model. All seven datasets were characterized by an uncertainty <0.3 on an annual scale. During 1982–2011, the mean global relative uncertainties followed the order LAI3g (8.5%)<GLASS (11.7%)<TCDR (13.0%)<GLOBMAP (19.6%) (Table 2). These statistics were comparable to the ancillary LAI products, with the order GEOV1 (6.4%)

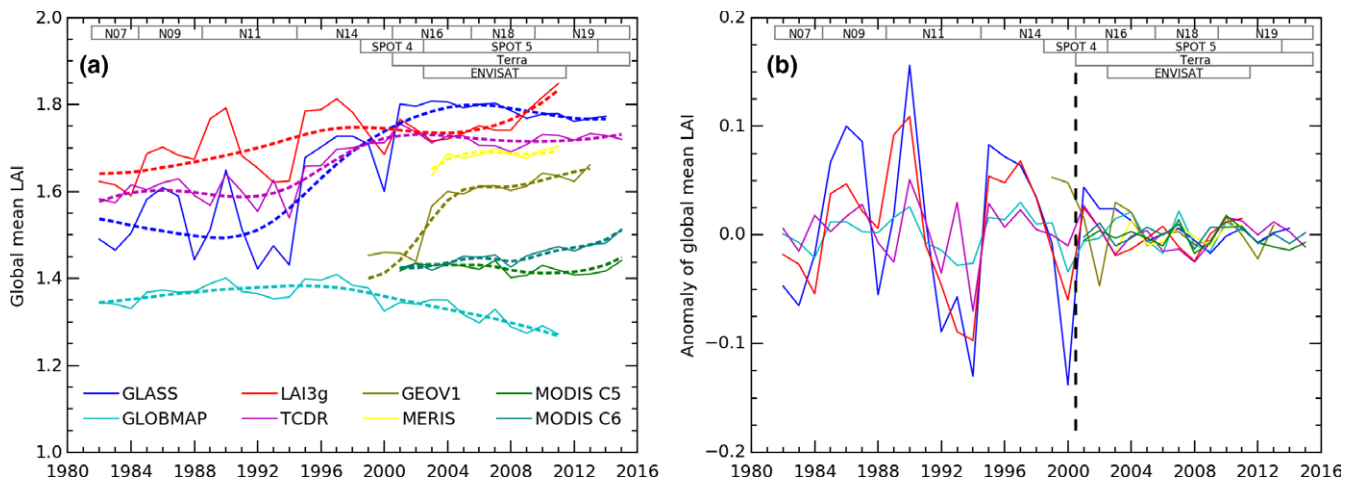


FIGURE 3 Global mean LAI, adaptive trends detected by the EEMD method, and detrended anomalies ($\text{m}^2 \text{m}^{-2} \text{year}^{-1}$). Anomalies (solid curves in b) are the subtraction of adaptive trends (dashed curves in a) from global mean LAI values (solid curves in a). Lifespans of different satellite platforms are also illustrated [Colour figure can be viewed at wileyonlinelibrary.com]

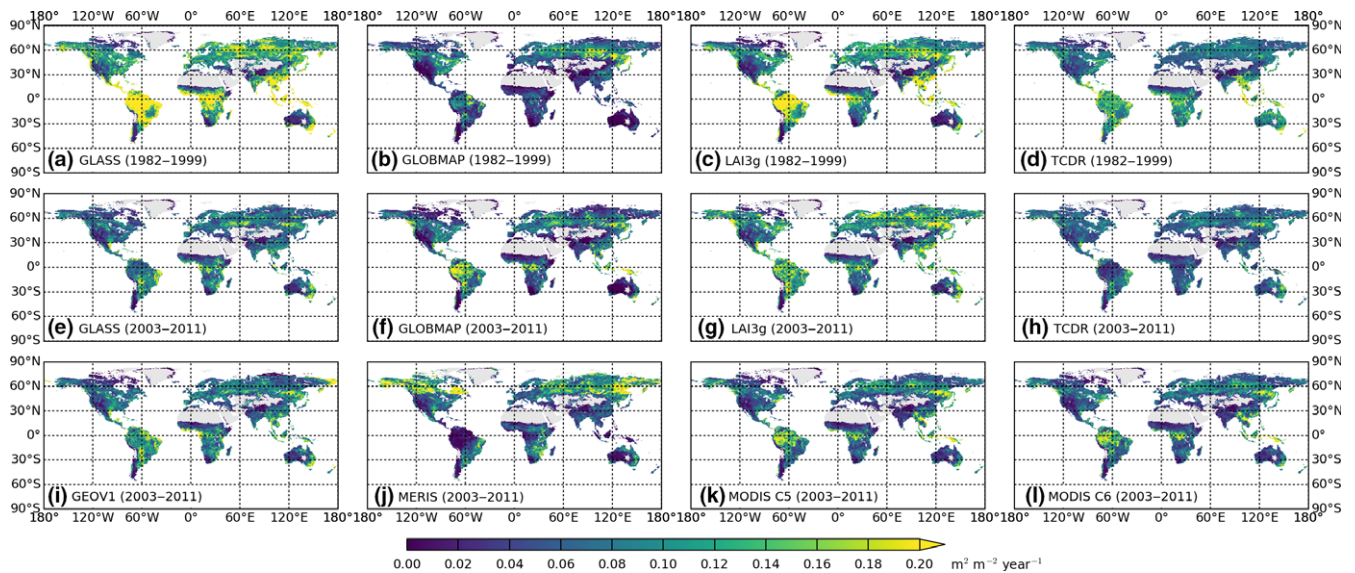


FIGURE 4 Interannual variability maps of growing season mean LAI for different LAI products during different time periods [Colour figure can be viewed at wileyonlinelibrary.com]

<MODIS C6 (13.6%)<MERIS (14.0%). Pre-MODIS (1982–2001) and overlap (2003–2011) periods showed only slight differences for GLOBAMP, LAI3g, and TCDR. In contrast, GLASS showed a notable decrease in 2001, resulting in the largest relative uncertainty variation (3.5%), while those of the other three LAI products were not larger than 2.0%.

The average maps of annual growing season mean LAI uncertainties and relative uncertainties were shown in Figure 6. The spatial pattern of uncertainty was generally in accordance with the distribution of LAI (Figure 6a). Tropical evergreen broadleaf forest regions showed the highest uncertainties (>0.40). Intermediate uncertainty values (0.20–0.30) were found for the subtropical and boreal forest regions. Uncertainties were low (<0.15) over sparsely

vegetated areas. Conversely (Figure 6b), low LAI regions including high latitude areas, Western America, Central Asia, Sub-Saharan Africa, Southern South America, Southern Africa, and Central and Western Australia generally showed large relative uncertainties (>20%). The relative uncertainties for forest regions varied between 10% and 20%. The spatial pattern of uncertainty variation was similar to the uncertainty map (Figure 7a), with high LAI regions showing larger variations over the 30-year period. Tropical forest regions showed the largest variations (>0.10), followed by subtropical and boreal forests (0.05–0.10). Meanwhile, the relative uncertainty variation showed similar spatial patterns to the relative uncertainty (Figure 7b), with large relative uncertainty variation mainly appearing in shrublands and grasslands.

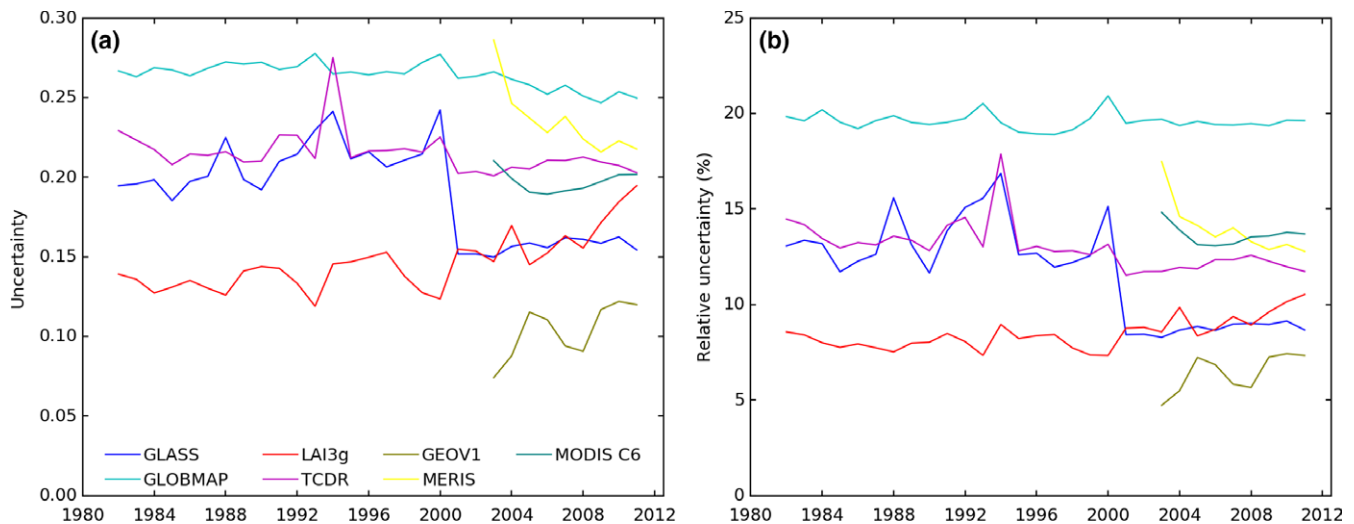


FIGURE 5 Uncertainties and relative uncertainties of global mean LAI for different LAI products [Colour figure can be viewed at wileyonlinelibrary.com]

4 | DISCUSSION

4.1 | Are current long-term LAI products intraconsistent over time?

This study revealed the intrainconsistency of the four long-term LAI products from two aspects: changes in interannual variability between two periods (1982–1999 and 2003–2011) and uncertainty variation over the 30-year period (1982–2011).

Remarkably different annual anomaly oscillations were observed before and after 2001 (Figure 3b). Such pronounced differences were not observed from other long-term datasets related to LAI, such as solar radiation (Sanchez-Lorenzo et al., 2015), temperature and precipitation (Yan et al., 2013), and CO₂ growth and carbon sink (Keenan et al., 2016). With the assumption that interannual variability was constant over the 30 years, we could use the ratio of interannual variability during 1982–1999 to that during 2003–2011 as an intraconsistency index. A larger deviation away from 1 would indicate a greater degree of intrainconsistency. Figure 8a showed that the global ratio followed the order GLOBMAP (1.2) < TCDR (2.1) < LAI3g (4.3) < GLASS (6.7), indicating the order of intrainconsistency at global scale. The most significant intrainconsistency appeared in tropical evergreen broadleaf forests (Figure 4a–d against Figure 4e–h), where the intrainconsistency index of TCDR, LAI3g, and GLASS was in range 6–12 (Figure 8a). High latitude (such as boreal evergreen needleleaf forests) and arid (such as woody savannas and grasslands) regions generally showed relatively low (<4) intrainconsistency (Figure 8a).

Figure 5 showed small and stable uncertainties and relative uncertainties of global annual growing season mean LAI for different products over the 30 years. Statistics listed in Table 2 showed that for all four long-term LAI products, the mean relative uncertainties were smaller than the 20% target accuracy and the relative uncertainty variations were smaller than the 10% target stability proposed

by GCOS (GCOS, 2011). However, we were unable to deduce they satisfied the requirements for accuracy or stability because the definitions in GCOS resemble both systematic error and random error, while uncertainty estimated from collocation error model approximates the latter (Fang, Wei, Jiang, et al. 2012; Fang, Wei, Liang, 2012). Instead, considering the unreasonably large oscillations before 2001 (Figure 3b), we might infer substantial relative uncertainty variations, i.e., instability, of systematic error component which were not captured by those of random error used in this study. We also noted that even relative uncertainty variation in random errors was highly correlated with the intrainconsistency index (Figure 8a), indicating the same order of intrainconsistency index at global scale: GLOBMAP (1.1%) < TCDR (1.6%) < LAI3g (2.0%) < GLASS (3.5%). Differing from intrainconsistency index, sparsely vegetated areas tended to have larger relative uncertainty variations as mean LAI was used as a normalization factor.

Satellite platform changes were likely a main reason for such intrainconsistencies. GLASS exhibited the most significant change in 2001 because of the change in satellite platform from NOAA/AVHRR to Terra/MODIS (Figure 3). NOAA/AVHRR itself was also multiplatform, composed of seven satellites and two sensors (AVHRR2 on board NOAA 07, 09, 11, and 14, and AVHRR3 on board NOAA 16, 18, and 19). This might have caused the intrainconsistency issue for LAI3g and TCDR to some extent. A satellite change also caused an abrupt change in the GEOV1 time series in 2003 (Fensholt, Rasmussen, Nielsen, & Mbow, 2009). Although GLOBMAP changed its platform in 2000 as well, it has exhibited relative good intraconsistency. This might be related to its special LAI retrieval algorithm, which divides the input into 10 bins, with SR–LAI relationship built within each bin. Such discretization strategy had been shown to be effective for improving the quality of retrievals from low-quality inputs and improving temporal smoothness (Liu et al., 2012). Cloudiness was another potential factor influencing year-to-year variations, especially for tropical evergreen forests which contribute most to global LAI interannual variability (Samanta et al., 2010). However, such

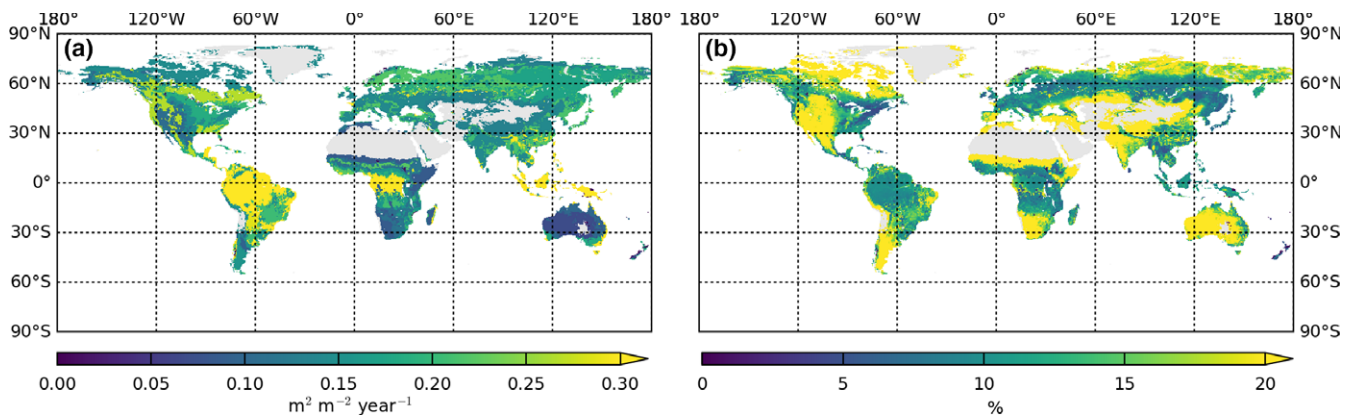


FIGURE 6 Average maps of (a) uncertainty and (b) relative uncertainty of growing season mean LAI for the four long-term LAI products [Colour figure can be viewed at wileyonlinelibrary.com]

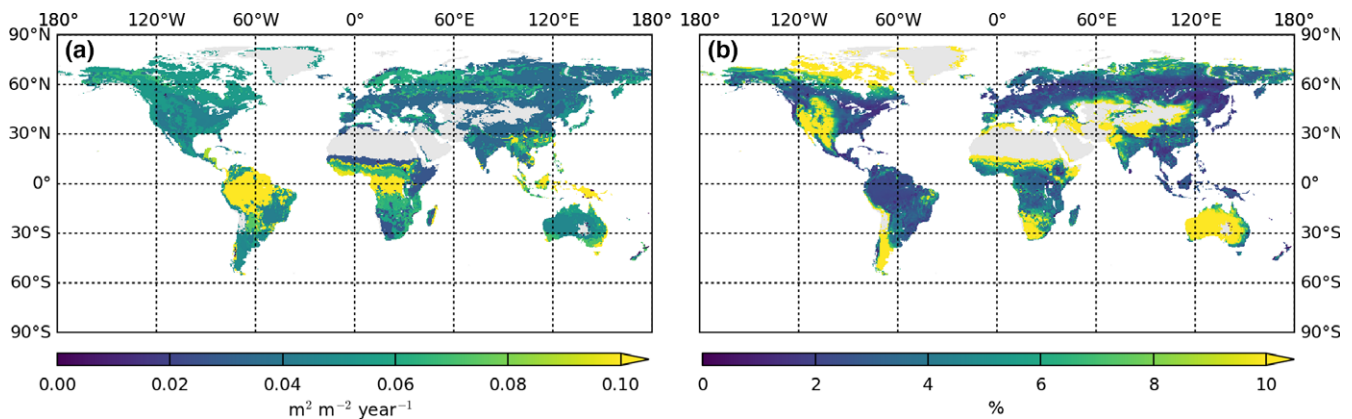


FIGURE 7 Average maps of (a) uncertainty variation and (b) relative uncertainty variation in growing season mean LAI for the four long-term LAI products [Colour figure can be viewed at wileyonlinelibrary.com]

difference was unlikely to cause decadal-level difference (the difference between 1982 and 1999 and 2003 and 2011) as long as the same cloud masking technique was used over the period.

4.2 | Are current long-term LAI products interconsistent with each other?

This study revealed the interinconsistency of the four long-term LAI products from discrepancies in trend and interannual variability for both global values and spatial patterns.

Greening or browning is a basic question of global change research (Myneni, Keeling, et al., 1997; Zhu et al., 2017), and the four long-term LAI products yielded different answers as different adaptive trend patterns were shown (Figure 3a). While adaptive trends revealed intrinsic properties of data associated with time scales, piecewise linear trends still provided intuitive quantitative intercomparison (Table 2). During the pre-MODIS period (1982–1999), all of the four long-term LAI products collectively detected the global greening trend (Figure 1), especially contributed by the widespread greening trends in the Northern Hemisphere (Figure 9a). However, their linear trends disagreed with each other in the following order: GLOBMAP ($0.026 \text{ m}^2 \text{ m}^{-2} \text{ decade}^{-1}$) < TCDR

(0.057) < LAI3g (0.082) < GLASS (0.115), and the linear trend maps showed generally different values between each other (Figure 2a–d). Note the global trends followed the same order as product intraconsistency. During the overlap period (2003–2011), the four long-term LAI products displayed different global trends that browning trends by GLASS and GLOBMAP, no trend by TCDR, and greening trend by LAI3g (Figure 1). Spatially, discrepancies in trend signs mainly appeared in tropical forests and temperate forests in Southeastern United States (Figure 9b and Figure 2e–h). In contrast, the three ancillary LAI products, GEOV1, MERIS, and MODIS C6 showed good agreements on trends both globally and spatially (Figures 1 and 2i–l).

Assessing environmental controls on vegetation variability is another essential issue in global change research (Jung, Reichstein, Schwalm, Huntingford, & Sitch, 2017; Seddon, Macias-Fauria, Long, Benz, & Willis, 2016), and the four long-term LAI products yielded considerably different annual anomalies (Figure 3b). During the pre-MODIS period (1982–1999), the interannual variability values followed the order GLOBMAP ($0.017 \text{ m}^2 \text{ m}^{-2} \text{ year}^{-1}$) < TCDR (0.027) < LAI3g (0.057) < GLASS (0.074). Note this was same as the order of the product intraconsistency. In addition, the agreement on spatial variations of annual anomalies was low, as was illustrated in Figure 10.

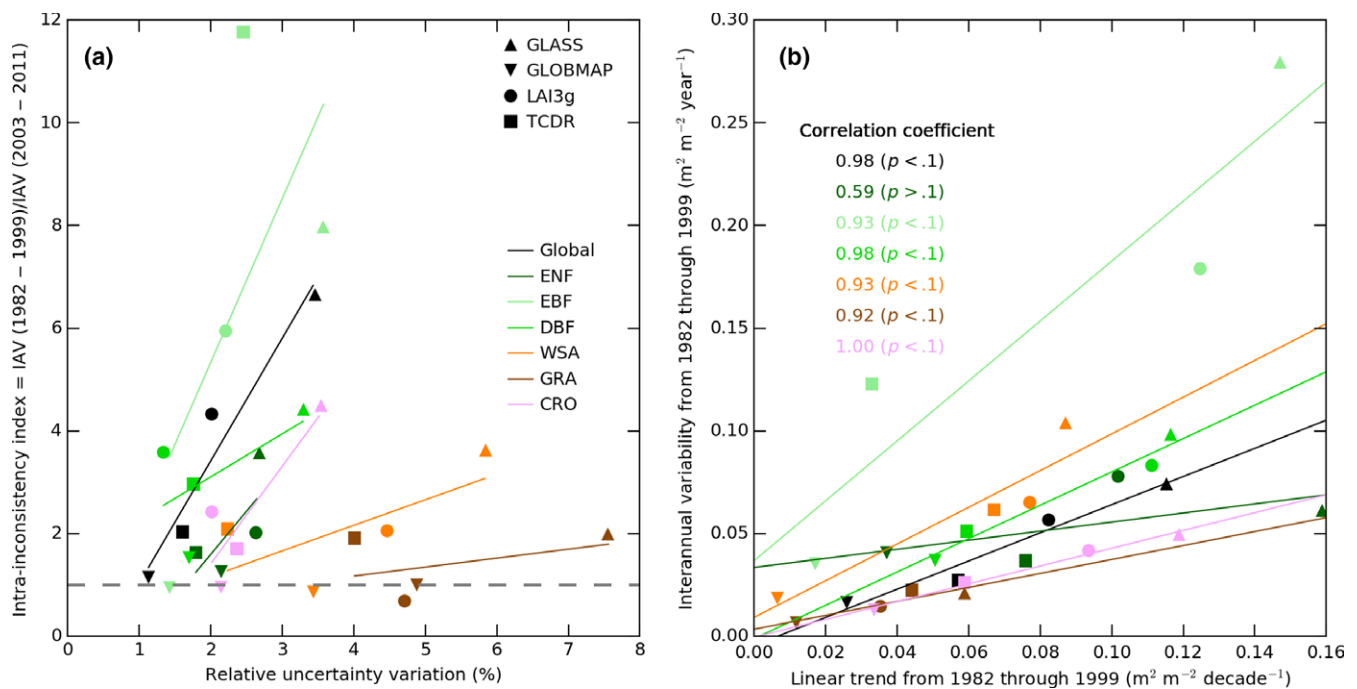


FIGURE 8 Linear relationships (a) between relative uncertainty variation over 30 years and the intra-inconsistency index (ratio of interannual variability (IAV) during 1982–1999 to that during 2003–2011), and (b) between linear trends and interannual variabilities during the pre-MODIS period (1982–1999), for global values and several big ecosystems. ENF: evergreen needleleaf forest; EBF: evergreen broadleaf forest; DBF: deciduous broadleaf forest; WSA: woody savanna; GRA: grasslands; CRO: croplands [Colour figure can be viewed at wileyonlinelibrary.com]

About 76% and 60% of pixels showed mean $R^2 < .40$, and relatively higher R^2 values were mainly observed in sparsely vegetated regions. Densely vegetated regions with high interannual variability but low correlation might be a severe problem for application. Recent studies indicated that a variety of remote sensing models could only explain less than 40% interannual variability in gross primary productivity (GPP) (Jiang & Ryu, 2016; Yuan et al., 2014). We inferred that LAI or equivalent input data, such as fraction of photosynthetically active radiation (FPAR), may limit the capacity of remote sensing models to reliably capture interannual variability in GPP. We further noted that linear trend was highly correlated with interannual variability during the pre-MODIS period (Figure 8b). This indicated that inconsistency of interannual variability was coupled with that of linear trend, except for boreal evergreen needleleaf forest.

The interinconsistency between LAI products was attributed to different methods of treating the satellite orbit change and sensor degradation issues. During the pre-MODIS period (1982–1999), different AVHRR data sources were used to generate the four LAI products: GLASS used LTDR reflectance data; GLOBMAP used GIMMS NDVI data; LAI3g used GIMMS NDVI3g data; and TCDR LAI used TCDR reflectance data. The major difference between LTDR and TCDR reflectance was that the latter normalized reflectance at nadir view and 45° solar zenith angle (Claverie et al., 2016), while the former did not apply bidirectional reflectance distribution function (BRDF) correction (Pedelty et al., 2007). The major difference between GIMMS NDVI and NDVI3g was that the former

applied the EMD method to minimize varying solar zenith-angle effects introduced by orbital drift (Tucker et al., 2005), while the latter placed more emphasis on among-instrument AVHRR calibration (Pinzon & Tucker, 2014). Marked differences between GIMMS NDVI and NDVI3g were also reported by Guay et al. (2014). With the fact that GLASS had much larger interannual variability than TCDR, and LAI3g had much larger interannual variability than GLOBMAP, we inferred that the BRDF effect on AVHRR data could cause exaggerated annual oscillation and consequently unrealistically large interannual variability in LAI products. Because there was a clear linear relationship between linear trend values and interannual variability values during this period (Figure 8b), we hypothesized that the BRDF effect caused by orbit change was also related to the trend inconsistency of the four long-term LAI products. However, further investigation is needed to evaluate this hypothesis. During the overlap period, both GLASS and GLOBMAP used MODIS reflectance data as input and they exhibited opposite linear trends to LAI3g and TCDR. This may be because both GLASS and GLOBMAP used MODIS C5 reflectance, which suffered from sensor degradation problems over the past decade (Wang et al., 2012). Such negative bias caused by sensor degradation also propagated to the MODIS C5 LAI product (Figure 1), which has also been reported in other studies (Yan et al., 2016). Because sophisticated calibration procedure has been applied on MODIS C6 reflectance (Zhang, Song, Band, Sun, & Li, 2017), and therefore the MODIS C6 LAI product has already remedied this issue by using corrected reflectance data

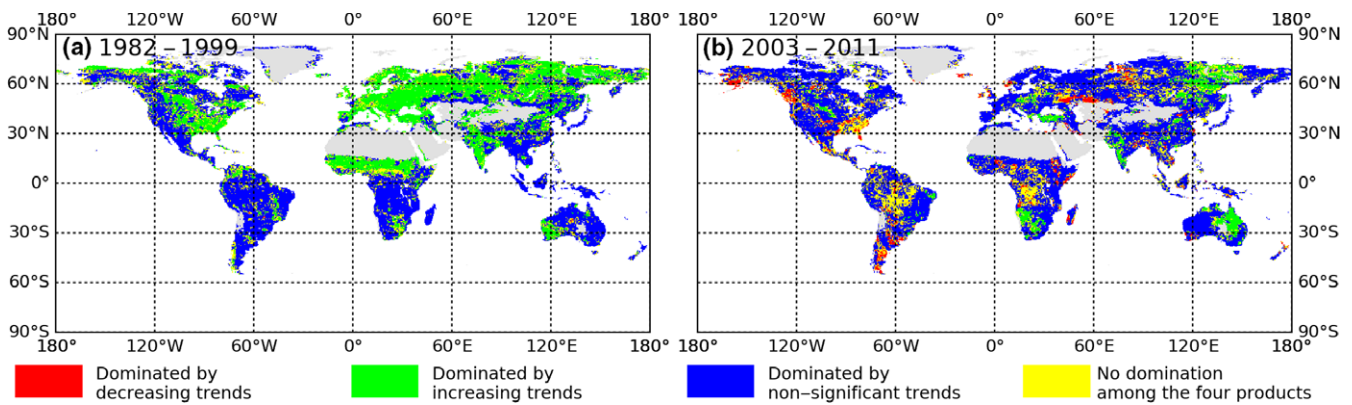


FIGURE 9 Agreement of GLASS, GLOBMAP, LAI3g, and TCDR on the signs of linear trends during (a) the pre-MODIS period (1982–1999) and (b) the overlap period (2003–2011). Domination refers to at least three of four products agreeing with each other. Significance: $p < .1$ [Colour figure can be viewed at wileyonlinelibrary.com]

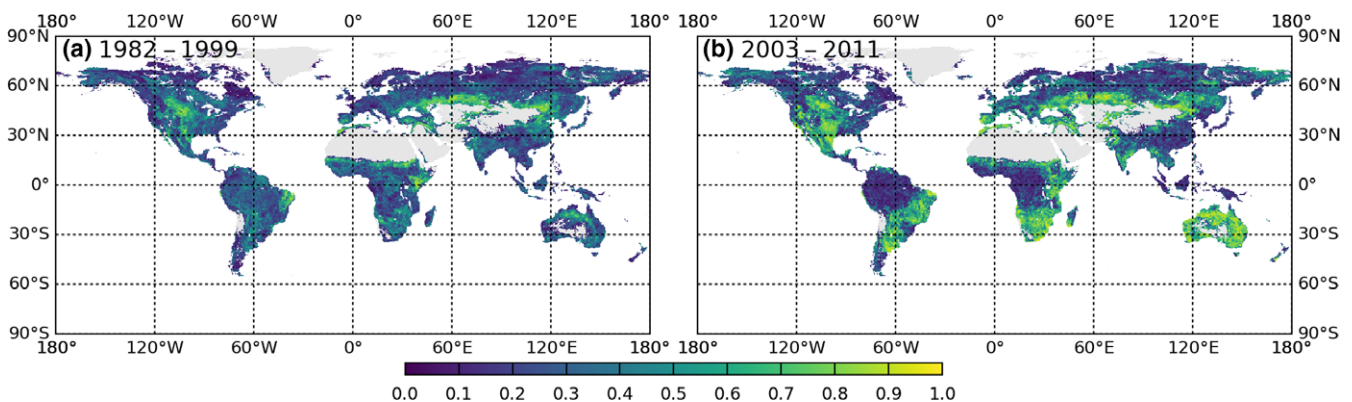


FIGURE 10 Mean R^2 maps between detrended anomalies of the four long-term LAI products during (a) the pre-MODIS period (1982–1999) and (b) the overlap period (2003–2011) [Colour figure can be viewed at wileyonlinelibrary.com]

(Figure 1), we suggest that the LAI product teams should also update their products using MODIS C6 reflectance or NDVI.

4.3 | Suggestions for the usage of current long-term LAI products

According to the uncertainty variation over the 30 years and the ratio of interannual variability during 1982–1999 to that during 2003–2011, GLOBMAP was the most intraconsistent product, followed by TCDR and LAI3g, whereas GLASS was the most inconsistent. Pronounced inconsistency was identified between these LAI products, and both linear trend and interannual variability during pre-MODIS period (1982–1999) were correlated with intraconsistency. However, recommendation of best LAI product during that period could not be drawn. Because scarce long-term continuous-field LAI measurements were available before 2000, it was impractical to evaluate annual trends and anomalies of the four long-term LAI products through direct validation using ground data. Further evaluations are still needed through indirect comparisons with climate data and land surface model simulations. Therefore, we suggest careful usage of current long-term LAI products. A single product or

even ensemble of the four products may not be directly considered as relevant references when interpreting long-term global carbon and water cycle studies during pre-MODIS period (1982 – 1999), except for several consensus: (i) trend signs, (ii) interannual variability values in high latitude regions, and (iii) annual anomalies in arid regions. During overlapped period (2003–2011), none of the four long-term LAI products agree well with the three reference LAI products (GEOV1, MERIS, and MODIS C6) in term of trend, but their interannual variabilities are relatively consistent with each other.

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MODIS global satellite LAI products. The authors declare no conflict of interest. English proofread was supported by Research Institute of Agriculture and Life Sciences at Seoul National University.

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SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

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